

3rd Year Undergraduate

Electronics & Communications

Engineering Department

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Faculty of Engineering

DESIGN AND IMPLEMENTATION OF A

Digitally Controlled Low-Side Buck Converter Prototype

with Closed-Loop PI Control

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Project Overview & Motivation

A Buck (step-down) DC-DC converter regulates a lower output voltage from a higher input. This project replaces analog compensation networks with fully digital, firmware-based PI control using an Arduino Nano (ATmega328P)— providing superior tunability, portability, and real-time adaptability without dedicated analog ICs.

Research Objective

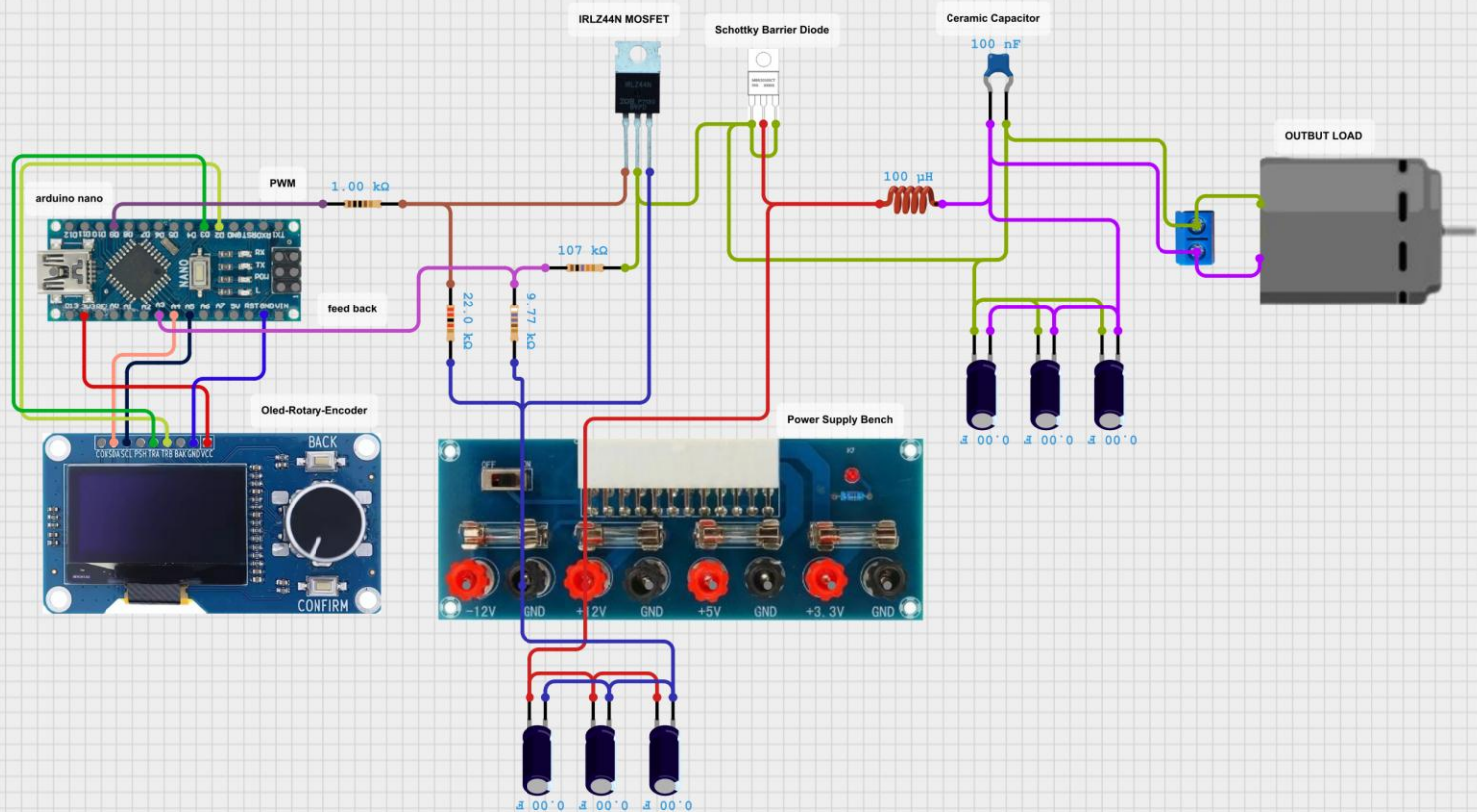
Achieve a continuously adjustable, regulated DC output (0–10.85 V) from an 11.85 V source, with measured steady-state error < 100 mV across the full output range.

Core Engineering Challenge

Implement accurate voltage sensing in a low-side switching topology where the MOSFET source is not at system ground — the so-called Floating Ground Problem.

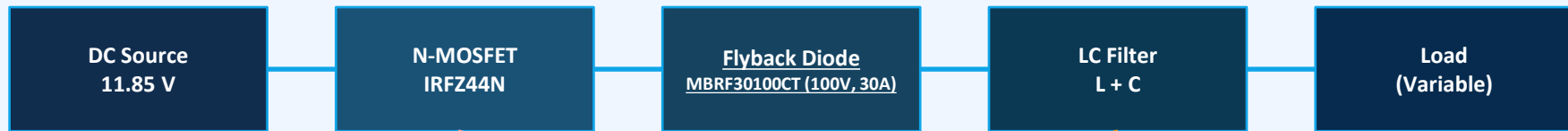
Key Contribution

Replace analog compensation with discrete PI control firmware, enabling real-time setpoint adjustment via rotary encoder with OLED display feedback at sub-300 ms refresh latency.

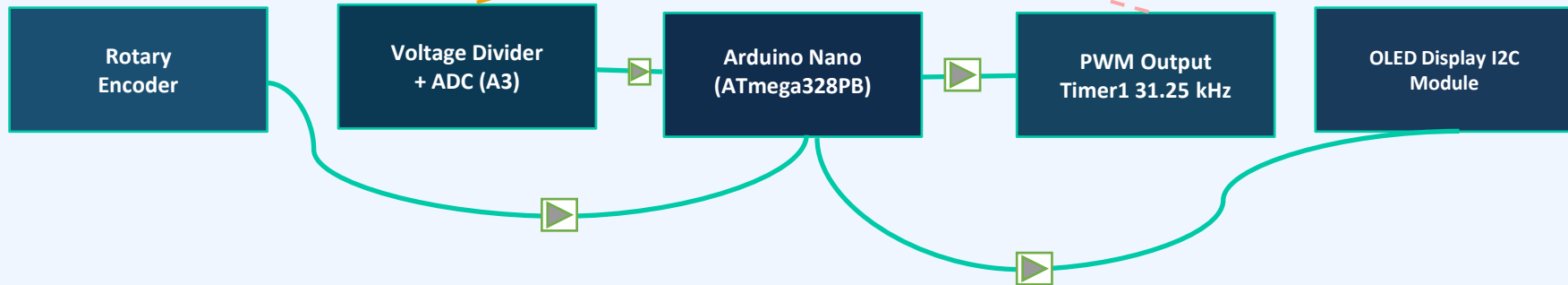


Complete System Architecture

▶ POWER PATH



CONTROL PATH (DIGITAL)



Power Flow

Control Signal

Feedback Path

Gate Drive

Hardware Topology: Low-Side N-Channel MOSFET Switching

N-Channel MOSFET (IRFZ44N) — Superior Rds(on)

N-channel devices exhibit significantly lower ON-state resistance ($R_{ds(on)} = 17.5\text{ m}\Omega$ @ $V_{gs} = 10\text{ V}$) compared to P-channel equivalents of the same die area, due to higher electron mobility. This directly minimises conduction losses ($P_{cond} = I_d^2 \times R_{ds(on)}$) and thermal dissipation at the switching element.

Low-Side Topology — Direct Gate Drive, No Bootstrap

Placing the MOSFET between the load return path and system ground (low-side configuration) allows the microcontroller's 5 V PWM output to drive the gate directly. A P-channel or high-side N-channel configuration would require a charge-pump gate driver or bootstrap capacitor circuit, adding BOM cost and PCB complexity.

Flyback (Freewheeling) Diode — MBRF30100CT Schottky

During the MOSFET OFF state, inductor current continuity demands a return path. The MBRF30100CT diode ($V_f = 0.84\text{ V}$ @ 15 A) provides this essential freewheeling path for the main inductor and motor, preventing destructive back-EMF spikes and maintaining Continuous Conduction Mode (CCM)."

Component Specifications	
Parameter	Value
Switch Device	IRFZ44N N-MOSFET
Rds(on)	17.5 mΩ @ Vgs=10V
Vds(max)	55 V
Id(max)	49 A
Vgs(th)	2–4 V
Freewheeling	MBRF30100CT Schottky
Vf (Schottky)	~0.84V @ 15A
Input Voltage	11.85 V
PWM Frequency	31.25 kHz
PWM Resolution	8-bit (0–255)

Differential Voltage Sensing: The Floating Ground Problem

Topology-Induced Measurement Constraint

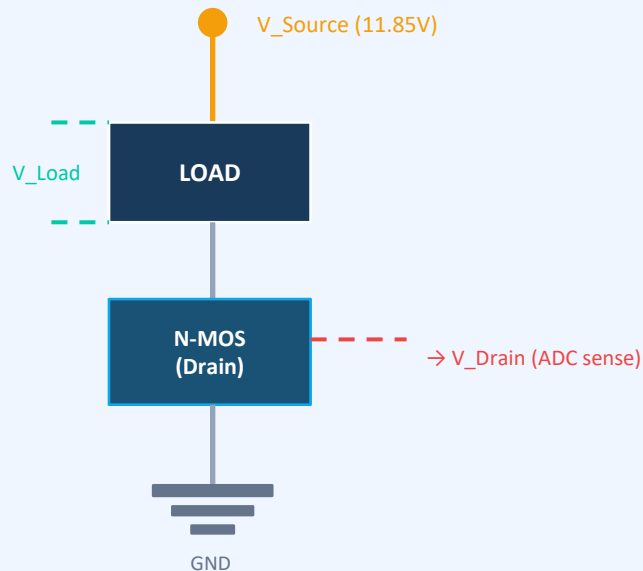
In the low-side topology, the MOSFET drain alternates between V_{Source} and V_{GND} on every switching cycle — it is NOT at a fixed DC potential. A direct ADC measurement of V_{Drain} therefore does not represent V_{Load} . Instead, the differential relationship across the load must be applied:

$$V_{\text{Load}} = V_{\text{Source}} - V_{\text{Drain}}$$

where V_{Drain} is sampled via the precision voltage divider at ADC pin A3

```
// readAndAdjust() – Differential Sensing Implementation  
float vDrain = (avgRaw * refVoltage / 1023.0) / dividerRatio;  
actualVoltage = (sourceVoltage - vDrain) * calFactor;  
if (actualVoltage < 0) actualVoltage = 0;
```

LOW-SIDE TOPOLOGY (Schematic)



ADC Precision Architecture: Three-Layer Strategy

01

Internal Bandgap Reference — $V_{ref} = 1.107 \text{ V}$ (Measured)

The ATmega328P's VCC rail is subject to $\pm 5\%$ variation ($\pm 250 \text{ mV}$), degrading ADC measurement integrity. The built-in bandgap reference (nominally 1.1 V, measured precisely at 1.107 V with a calibrated bench multimeter) is temperature-stable and VCC-independent. The resulting ADC LSB = $1.107 \text{ V} / 1023 \approx 1.08 \text{ mV}$ — a 4.5× improvement in LSB resolution versus the 5 V reference. Activated via: `analogReference(INTERNAL)`

02

Custom Voltage Divider — $R_1 = 107.3 \text{ k}\Omega$, $R_2 = 9.77 \text{ k}\Omega$, $\beta = 0.083454$

The divider ratio $\beta = R_2 / (R_1 + R_2) = 9.77 / (107.3 + 9.77) = 0.083454$ scales V_{Drain} (0–12 V) into the ADC input range (0–1.107 V). High-impedance resistors minimise shunt current loading on the power path. The exact measured ratio is embedded directly in firmware to account for 1% component tolerances, rather than relying on nominal design values.

03

Software Calibration Factor — $k = 0.92$ (Empirically Derived)

After hardware compensation, a residual systematic offset remained between computed and physically measured output voltages — attributable to unmeasured parasitic resistances in PCB traces, probe contact resistance, and inductor DCR. A multiplicative factor $k = (V_{\text{measured}} / V_{\text{computed}}) = 0.92$ was derived empirically and applied in firmware as a final correction stage.

$$V_{\text{Drain}} = (\text{avgRaw} \times 1.107 / 1023.0) / 0.083454 \quad \rightarrow \quad \text{actualVoltage} = (11.85 - V_{\text{Drain}}) \times 0.92$$

High-Frequency PWM: Timer1 Prescaler Configuration

Problem: Default 490 Hz Falls in the Audible Band

Arduino's default Timer1 PWM frequency (490 Hz) lies within the human audible range (20 Hz – 20 kHz). At this frequency, the energy storage inductor oscillates audibly under repetitive magnetic flux variations, producing acoustic noise — an unacceptable artefact. A lower frequency also demands a physically larger inductance for equivalent output ripple ($\Delta i_L = V_{out} \cdot (1-D) / (L \cdot f_{sw})$).

Solution: Prescaler = 1 → f_PWM = 31.25 kHz

Manipulating the TCCR1B clock-select bits (CS12:CS11:CS10 = 0:0:1) sets prescaler N=1, directly clocking Timer1 from the 16 MHz system oscillator. In phase-correct 8-bit PWM mode, the switching frequency is:

$$f_{PWM} = f_{CPU} / (2 \times N \times 256) = 16 \text{ MHz} / (2 \times 1 \times 256) = 31,250 \text{ Hz}$$

```
// setup() – Timer1 Register Manipulation (TCCR1B)
TCCR1B = (TCCR1B & 0b11111000) | 0x01; // CS10=1: prescaler
N=1
```

PRESCALER SELECTION MATRIX

Prescaler N	f_PWM	In Audible Band?
1024	~30 Hz	No (sub-audio)
256	122 Hz	No (sub-audio)
64 (default)	490 Hz	YES X
8	3.92 kHz	YES X
1 ← CHOSEN	31.25 kHz	No ✓

Signal Integrity & Noise Suppression Strategy

Hardware: Decoupling Capacitor Network

Root Cause: The switching MOSFET draws discontinuous (pulsed) current from the supply bus. Each transition generates a rapid di/dt transient. Via Lenz's law ($v = L \cdot di/dt$), parasitic PCB trace and wire inductance induces voltage spikes onto V_{Source} , corrupting both the ADC reference and voltage measurements.

Solution: Large bulk electrolytic capacitors (470–1000 μF) across the input bus act as a local charge reservoir, supplying transient current during switch-on and absorbing voltage spikes during switch-off. Small ceramic bypass capacitors (100 nF) at the ADC pin suppress switching harmonics above 1 MHz. Together, this maintains V_{Source} stability within ± 50 mV across switching cycles.

Eliminates supply-induced ADC measurement errors and prevents saturation during fast switching transients.

Firmware: Digital Oversampling (N = 100 Samples)

Theory: By averaging $N = 100$ independent ADC samples per control cycle, the random (white) noise component is attenuated by $\sqrt{N} \approx 10$. The SNR improvement equals $20 \cdot \log_{10}(10) = 20$ dB, equivalent to extending ADC resolution by $\log_2(10) \approx 3.32$ bits — yielding an effective resolution of ~ 13.3 bits under white-noise conditions (ENOB improvement via oversampling).

```
// 100-sample oversampling for noise rejection

long sum = 0;
for(int i = 0; i < 100; i++) {
    sum += analogRead(feedbackPin);
}
float avgRaw = sum / 100.0;
```

SNR improvement: +20 dB | Effective resolution: ~ 13.3 bits

Closed-Loop Control: Discrete PI Controller Implementation

$$\begin{aligned} \text{Continuous: } u(t) &= K_p \cdot e(t) + K_i \cdot \int e(t) dt \\ \text{Discrete: } u[n] &= K_p \cdot e[n] + K_i \cdot \sum (T_c \cdot e[k]) \end{aligned}$$

$K_p = 5.0$

Proportional Gain

Delivers immediate corrective action proportional to the instantaneous voltage error $e[n] = V_{\text{set}} - V_{\text{out}}$. Dominant corrective term for large deviations; tuned to achieve fast rise time without generating overshoot in the step response.

$K_i = 0.1$

Integral Gain

Accumulates error over time to eliminate steady-state offset — a fundamental limitation of proportional-only control. Conservatively sized at 0.1 to prevent integrator windup under rapid setpoint changes.

**± 50
Clamp**

Anti-Windup Guard

The error integral is hard-clamped to $[-50, +50]$ via `constrain()`. Prevents integrator windup — the pathological condition where a saturated actuator accumulates unbounded integral error, causing severe overshoot and extended recovery time.

**$\pm 0.05V$
Dead**

Deadband Control

PI action is suppressed and the integral is reset when $|e[n]| \leq 0.05 \text{ V}$. This prevents limit-cycle oscillation (hunting) — the high-frequency toggling of PWM output caused by noise-induced oscillation around the setpoint.

Firmware Architecture: Non-Blocking Cooperative Multitasking

The firmware employs a cooperative multitasking model using the `millis()` 32-bit hardware counter — eliminating blocking `delay()` calls that stall the MCU and prevent concurrent task execution. Each subsystem is assigned an independent update period, evaluated via timestamp comparison on every main loop iteration. This allows the control loop, display refresh, and encoder polling to execute at independent, decoupled rates.

Task	Period	Priority	Mechanism	Engineering Rationale
Rotary Encoder Decode	Every loop	Critical	Encoder lib + Hardware Interrupts	Quadrature decoding requires edge-sensitive counting on every main-loop pass; latency > 1 loop-pass risks missed pulses
PI Control Loop	60 ms	High	<code>millis()</code> timestamp	Control bandwidth ≈ 16.7 Hz — sufficient for slow output voltage dynamics; budget allows 100-sample ADC window
100-sample ADC Average	(within PI)	High	Blocking for-loop (~5 ms)	Nested within 60 ms control slot; 5 ms execution time does not violate system non-blocking contract
PWM Output Write	On PI update	Critical	<code>analogWrite()</code> → <code>Timer1</code>	Updated each PI iteration; reflects latest corrective action to hardware switch without additional latency
OLED Display Refresh	300 ms	Low	<code>millis()</code> timestamp	Above human flicker threshold; reduces I ² C bus contention and avoids starving the time-critical control loop

** Interrupt-driven quadrature decoding provided by the Encoder library (Paul Stoffregen). The 100-sample ADC loop (~5 ms) is a controlled blocking segment nested within the 60 ms control budget and does not violate the system-level non-blocking contract.*

System Specifications & Design Parameters

Power Stage	
Input Voltage (V_in)	11.85 V
Output Range	0 – 10.85 V
Switching Device	IRFZ44N N-MOSFET
Rds(on)	17.5 mΩ
Freewheeling Diode	MBRF30100CT Schottky
Diode Vf	~0.84 V @ 15 A
PWM Frequency	31.25 kHz
PWM Bit Depth	8-bit (0–255)

Sensing & ADC	
ADC Reference	Internal Bandgap
V_ref (Measured)	1.107 V
Divider Ratio (β)	0.083454
R ₁ (Top)	107.3 kΩ
R ₂ (Bottom)	9.77 kΩ
Cal. Factor (k)	0.92
Oversampling (N)	100 samples
Effective Resolution	~13.3 bits

Digital Control (PI)	
Controller Type	Discrete PI
Proportional Gain Kp	5.0
Integral Gain Ki	0.1
Integral Clamp	±50.0
Deadband Threshold	±0.05 V
Control Period (Tc)	60 ms
Control Rate	~16.7 Hz
Display Update	300 ms

CONCLUSION

What Was Achieved

- Regulated adjustable output (0–10.85 V) with < 100 mV steady-state error
- N-channel low-side gate drive directly from 5V MCU — no auxiliary ICs
- Floating ground differential sensing with calibrated precision divider
- 31.25 kHz PWM via Timer1 register manipulation — zero audible noise
- PI control with anti-windup clamping and deadband — stable under transients
- Non-blocking cooperative firmware: parallel control (60 ms) + UI (300 ms)

Future Work & System Upgrades

Current Sensing & Overcurrent Protection

Integration of an inline shunt resistor or Hall-effect sensor (ACS712) enables an inner current control loop, output current limiting, and short-circuit protection — mandatory for any production-grade power supply design.

Synchronous Buck Topology (Active Rectification)

Replace the MBRF30100CT Schottky freewheeling diode with a synchronous N-MOSFET switch. Eliminates the $V_f \approx 0.84$ V conduction loss, improving efficiency at high duty cycles. Requires precise dead-time management to prevent shoot-through.

PID Controller + Kalman Filter / State Observer

Add a derivative term (K_d) with a first-order IIR low-pass filter to suppress noise amplification inherent to differentiation. A discrete Kalman filter or state-space observer would further improve transient load rejection.

Phase II: Productization

Future Roadmap

- ⚡ **ATX Integration:** Repurposing PC supplies for high-current 12V/5V rails.
- 🔌 **Buck-Boost Upgrade:** Transitioning to a 0V - 36V variable range.
- 📦 **3D Printed Enclosure:** Professional benchtop housing with active cooling.
- 🖨️ **Custom PCB:** Custom-routed boards for minimized parasitic noise.

